

Student_performance

July 25, 2025

STUDENT PERFORMANCE IN EXAMS-DATA ANALYSIS

Goal:

To analyze how student demographics, parental education, lunch status, and test preparation impact students performance in math, reading, and writing, and to identify key factors that can help improve student scores.

Questions to be asked:

1. What is the gender distribution of students?
2. What is the average math score for males vs females?
3. What is the average reading score for males vs females?
4. What is the average writing score for males vs females?
5. How does parental level of education affect average math scores?
6. Do students who complete the test preparation course have higher average math scores?
7. What is the distribution of students by race/ethnicity?
8. What is the distribution of parental levels of education?
9. How many students have standard vs free/reduced lunch?
10. Do students who completed the test preparation course score higher on average?
11. How does parental education level affect average math/reading/writing scores?
12. How does lunch type affect scores?
13. What is the relationship between math and reading scores?
14. What is the relationship between math and reading scores?
15. Do average scores differ between male and female students?

I. IMPORT THE LIBRARIES

II. LOAD THE DATA

	gender	race/ethnicity	parental level of education	lunch	\
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	

3	male	group A	associate's degree	free/reduced
4	male	group C	some college	standard
..	...	...	...	...
995	female	group E	master's degree	standard
996	male	group C	high school	free/reduced
997	female	group C	high school	free/reduced
998	female	group D	some college	standard
999	female	group D	some college	free/reduced

	test preparation course	math score	reading score	writing score
0	none	72	72	74
1	completed	69	90	88
2	none	90	95	93
3	none	47	57	44
4	none	76	78	75
..	...	...	...	...
995	completed	88	99	95
996	none	62	55	55
997	completed	59	71	65
998	completed	68	78	77
999	none	77	86	86

[1000 rows x 8 columns]

III. UNDERSTAND THE DATA

	gender	race/ethnicity	parental level of education	lunch \
0	female	group B	bachelor's degree	standard
1	female	group C	some college	standard
2	female	group B	master's degree	standard
3	male	group A	associate's degree	free/reduced
4	male	group C	some college	standard

	test preparation course	math score	reading score	writing score
0	none	72	72	74
1	completed	69	90	88
2	none	90	95	93
3	none	47	57	44
4	none	76	78	75

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1000 entries, 0 to 999
```

```
Data columns (total 8 columns):
```

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	gender	1000 non-null	object
1	race/ethnicity	1000 non-null	object
2	parental level of education	1000 non-null	object

```

3   lunch                                1000 non-null  object
4   test preparation course              1000 non-null  object
5   math score                          1000 non-null  int64
6   reading score                       1000 non-null  int64
7   writing score                        1000 non-null  int64

```

dtypes: int64(3), object(5)

memory usage: 62.6+ KB

None

	math score	reading score	writing score
count	1000.000000	1000.000000	1000.000000
mean	66.08900	69.169000	68.054000
std	15.16308	14.600192	15.195657
min	0.00000	17.000000	10.000000
25%	57.00000	59.000000	57.750000
50%	66.00000	70.000000	69.000000
75%	77.00000	79.000000	79.000000
max	100.00000	100.000000	100.000000

```

Index(['gender', 'race/ethnicity', 'parental level of education', 'lunch',
      'test preparation course', 'math score', 'reading score',
      'writing score'],
      dtype='object')

```

```
array(['female', 'male'], dtype=object)
```

```
array(['group B', 'group C', 'group A', 'group D', 'group E'],
      dtype=object)
```

```
array(["bachelor's degree", 'some college', "master's degree",
      "associate's degree", 'high school', 'some high school'],
      dtype=object)
```

```
array(['standard', 'free/reduced'], dtype=object)
```

```
array(['none', 'completed'], dtype=object)
```

```
np.int64(0)
```

IV. CLEAN THE DATA

```

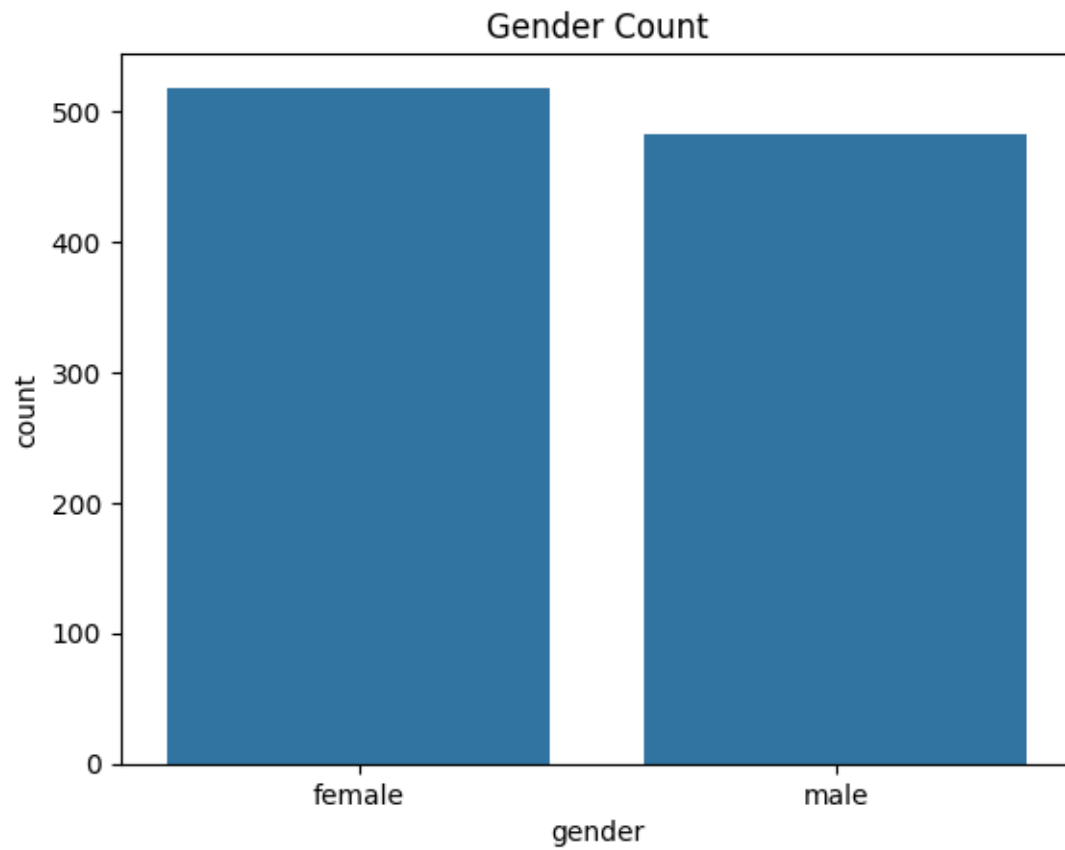
gender                                0
race/ethnicity                        0
parental level of education           0
lunch                                 0
test preparation course                0
math score                            0
reading score                         0

```

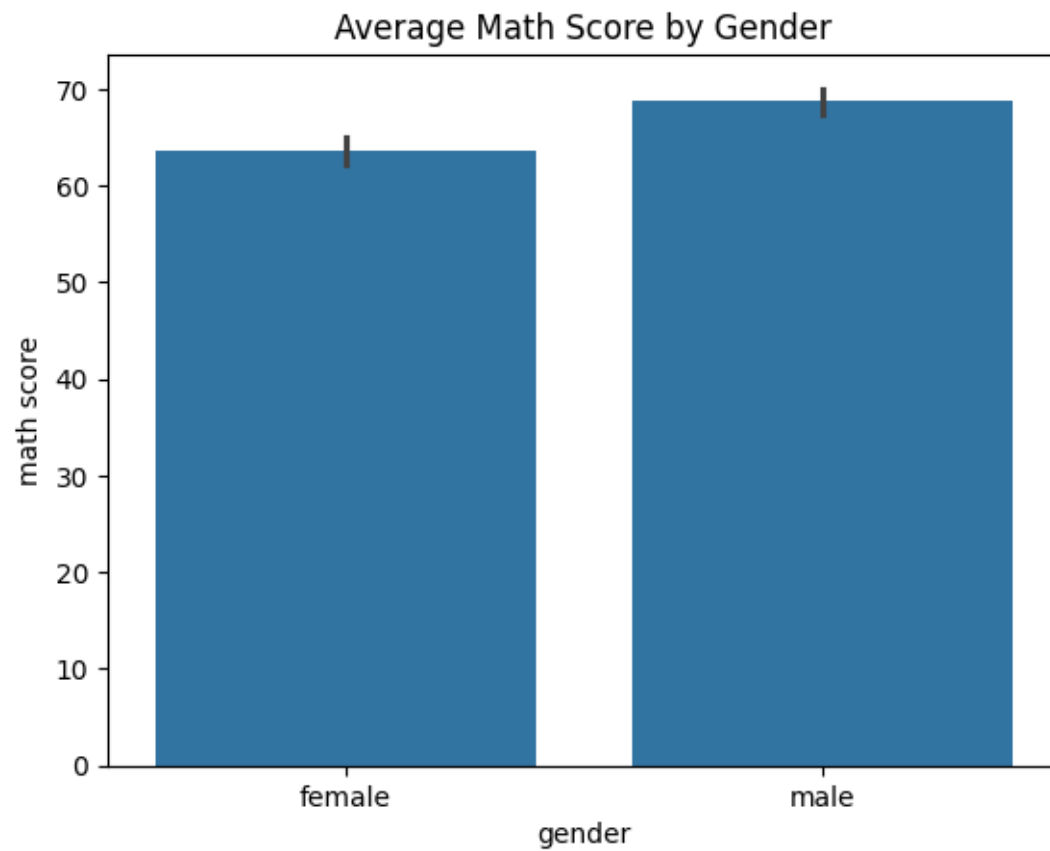
```
writing score      0  
dtype: int64
```

V. VISUALIZE

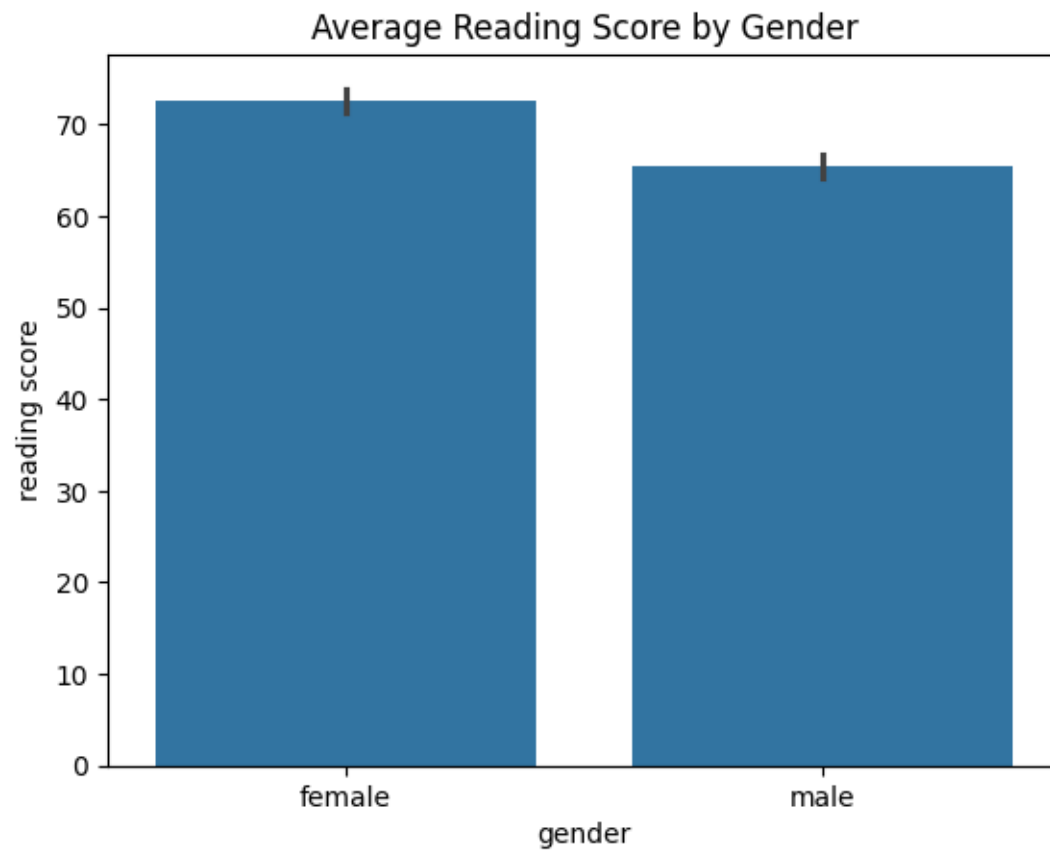
1. What is the gender distribution of students?



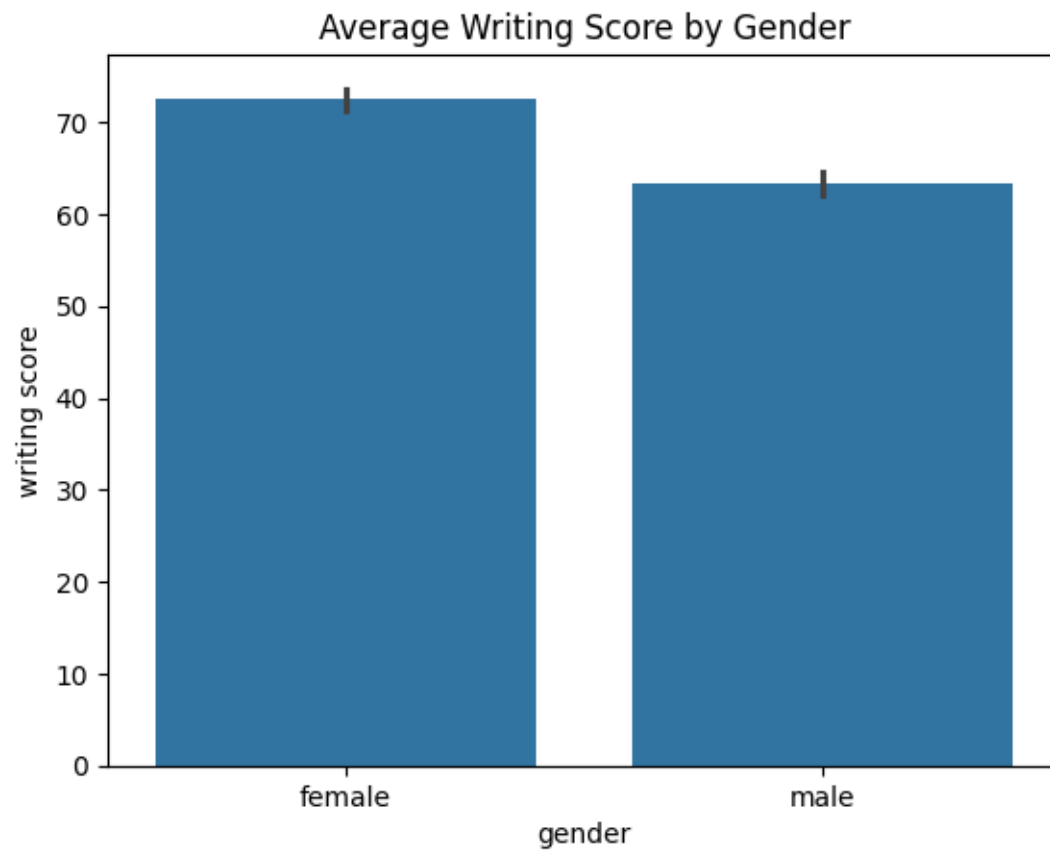
2. What is the average math score for males vs females?



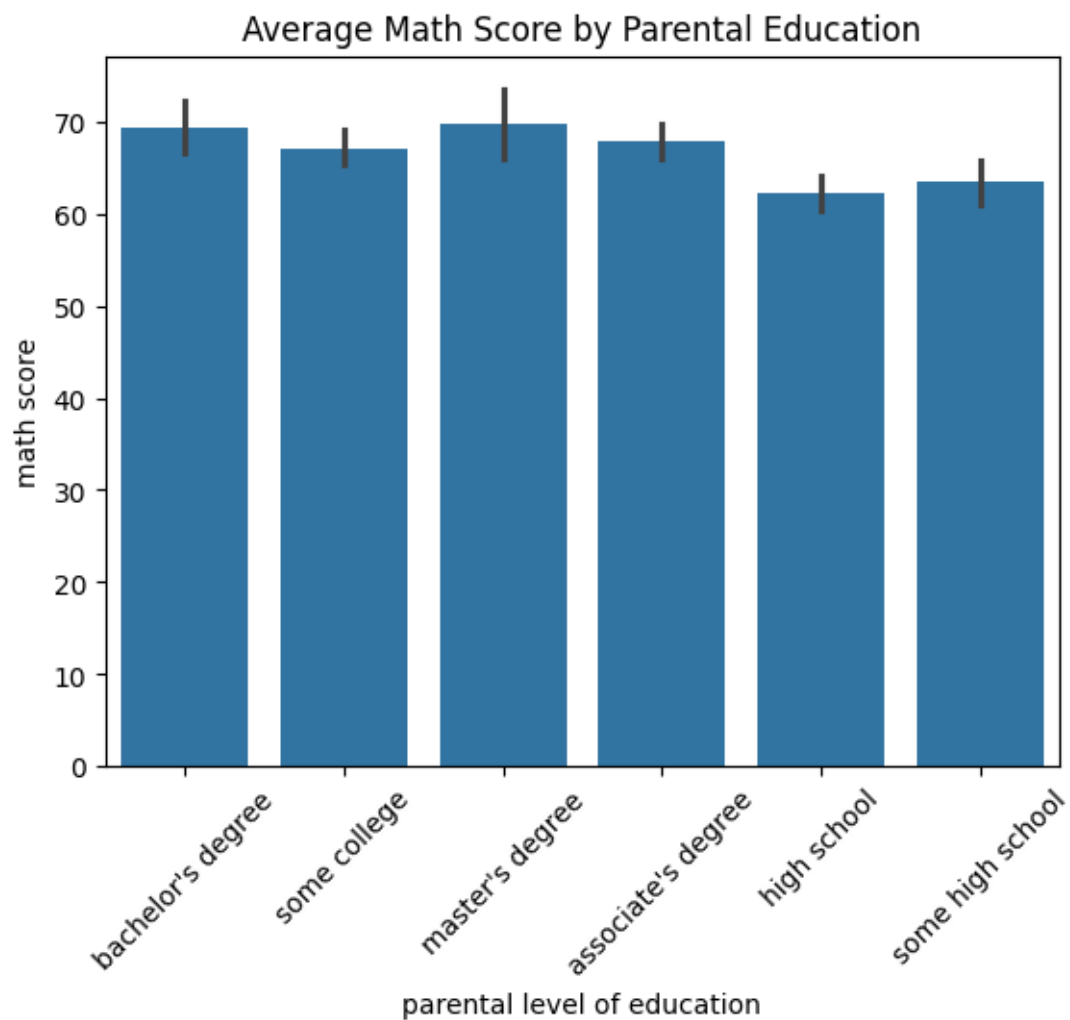
3. What is the average reading score for males vs females?



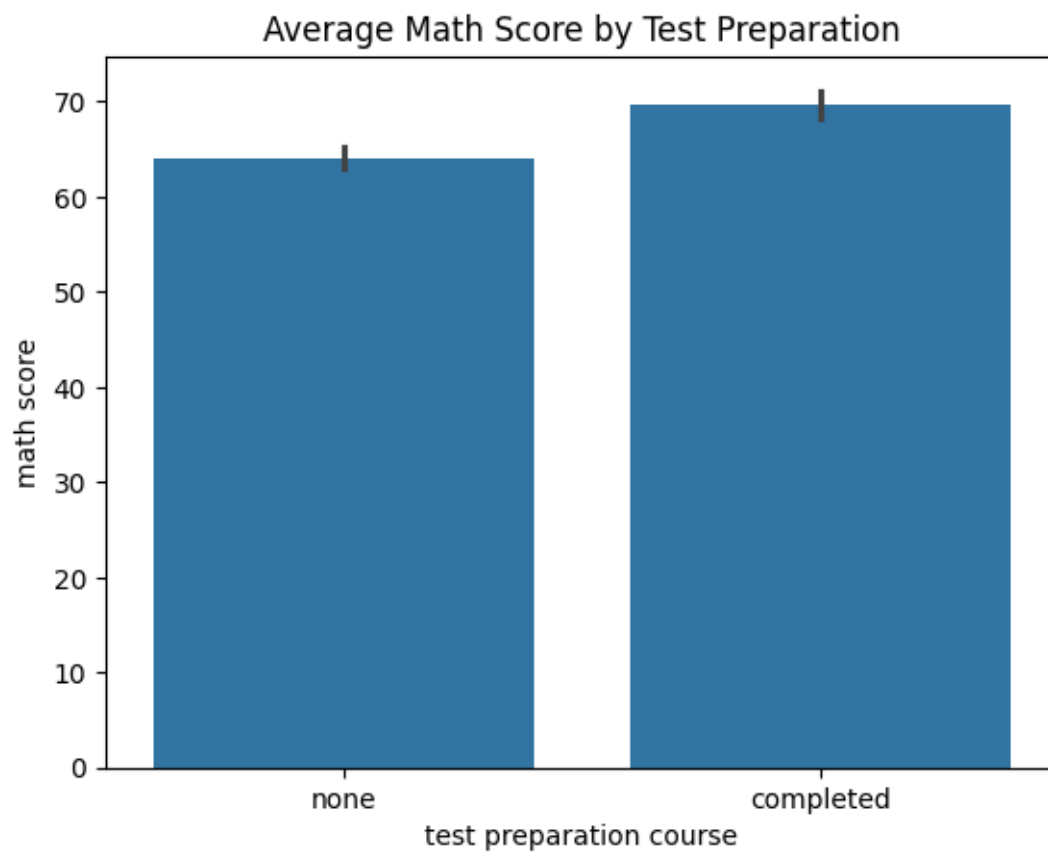
4. What is the average writing score for males vs females?



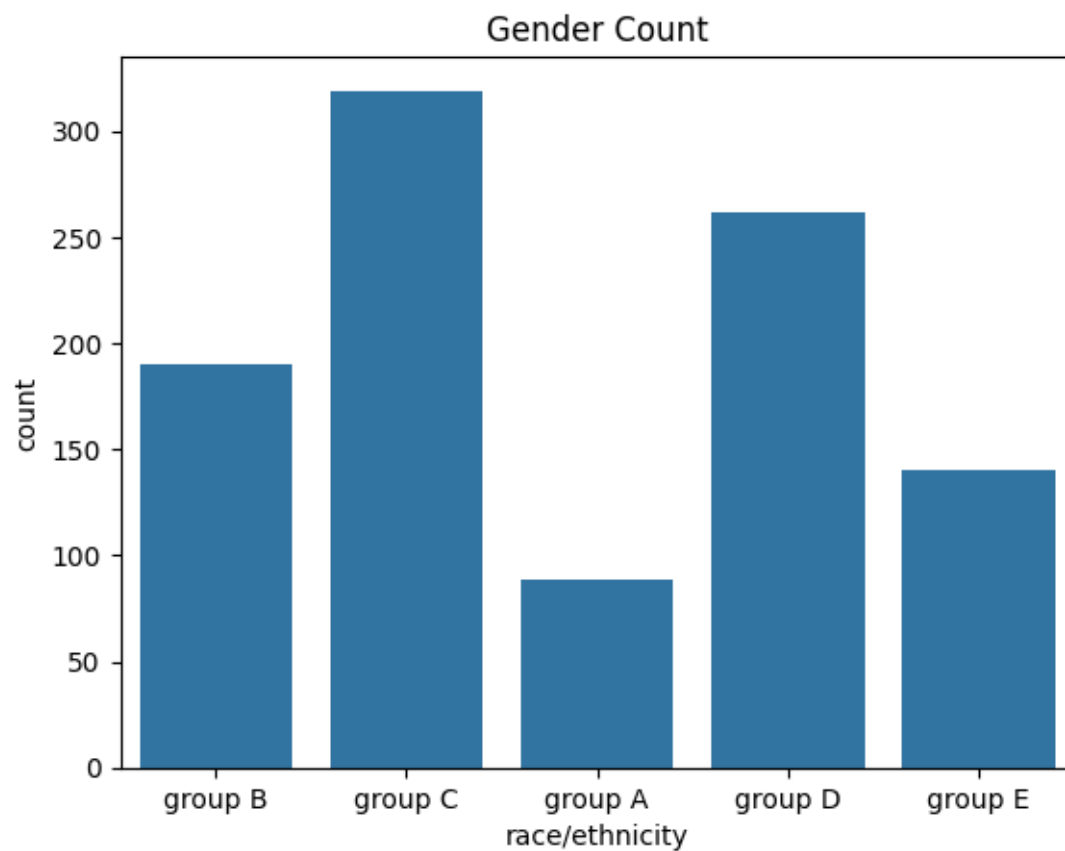
5. How does parental level of education affect average math scores?



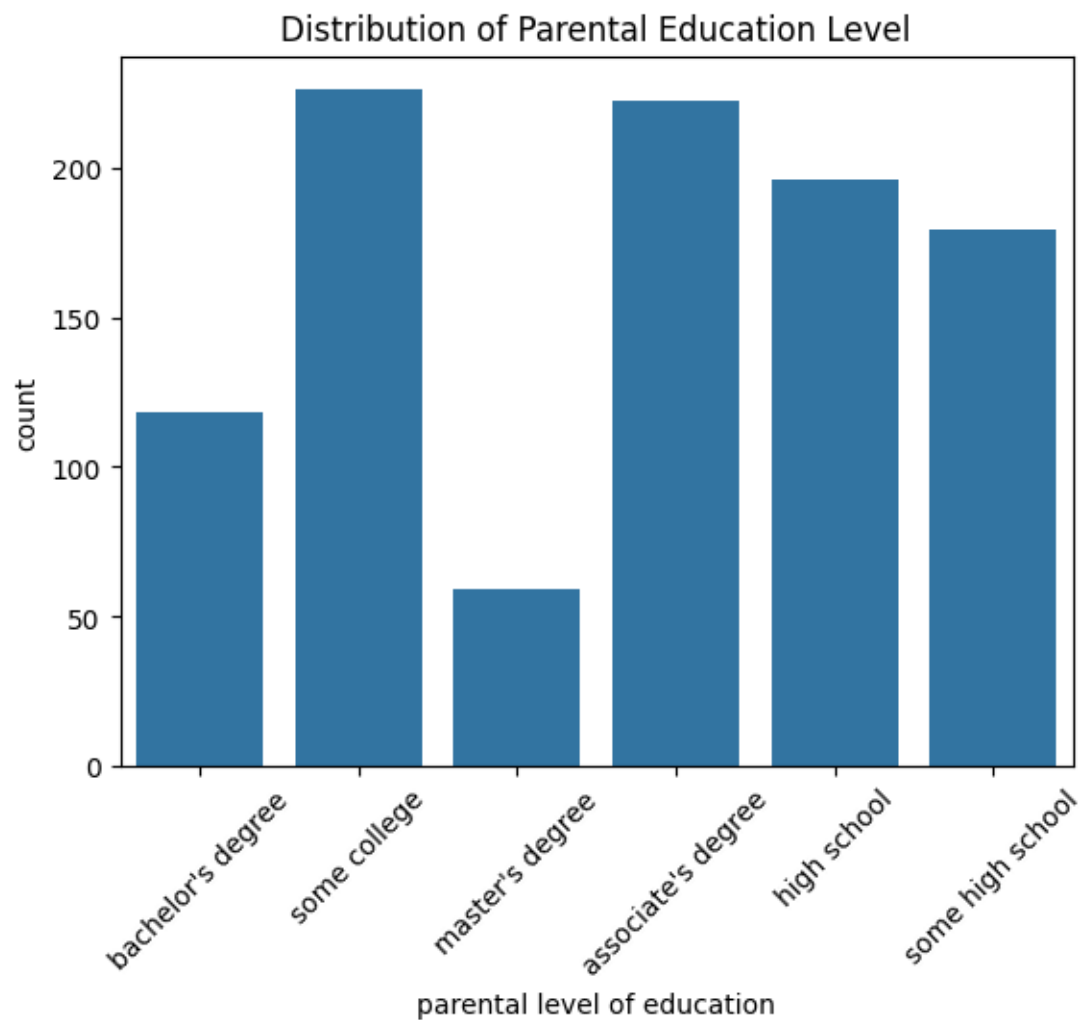
6. Do students who complete the test preparation course have higher average math scores?



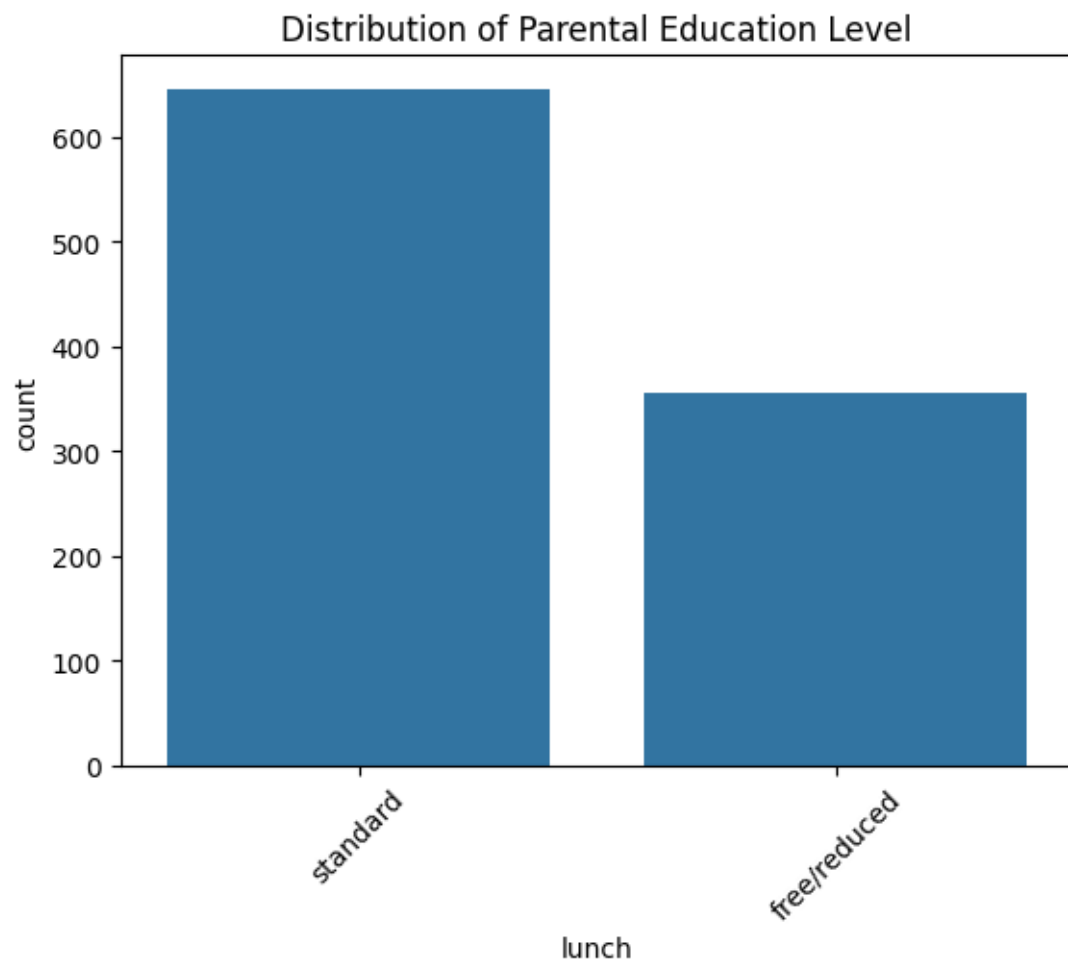
7. What is the distribution of students by race/ethnicity?



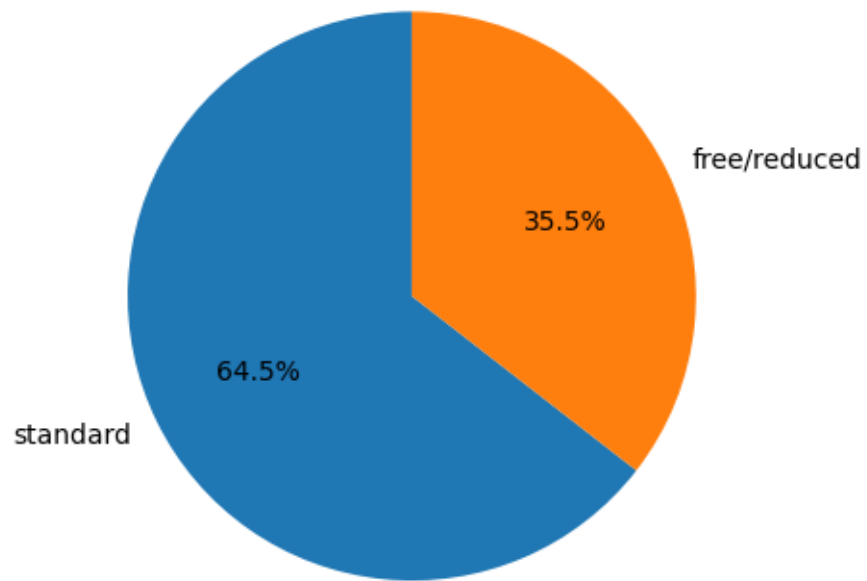
8. What is the distribution of parental levels of education?



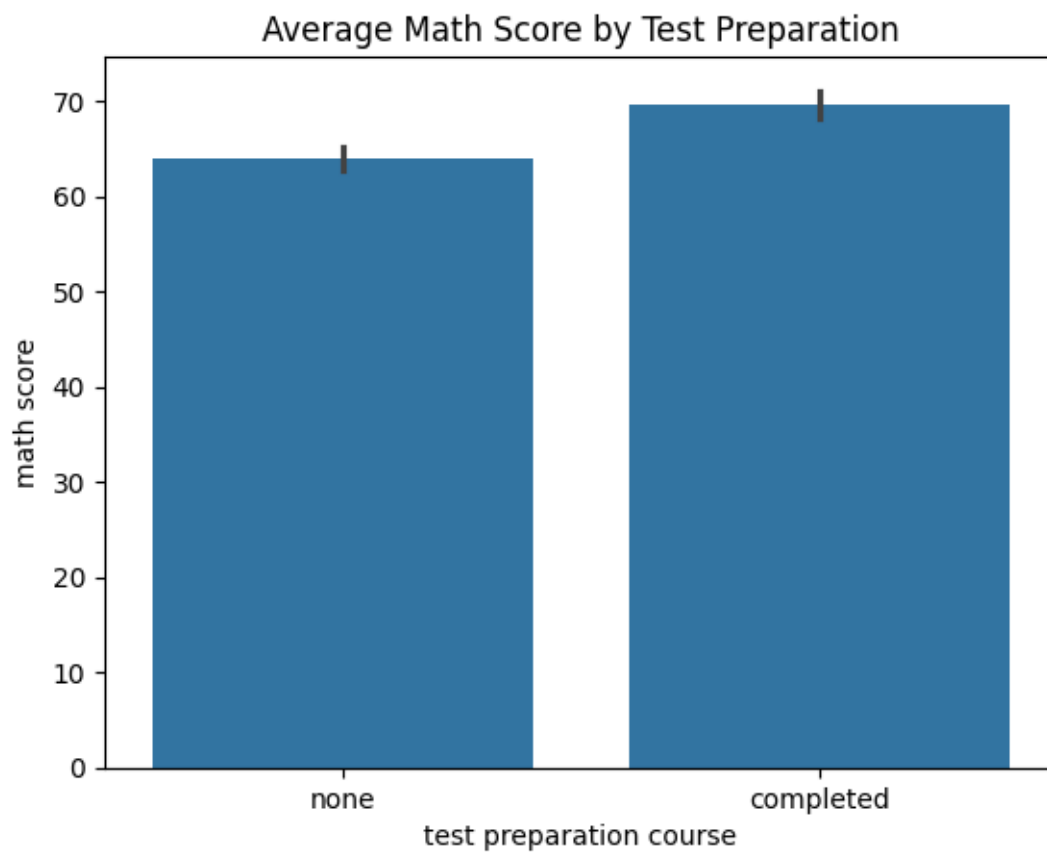
9. How many students have standard vs free/reduced lunch?



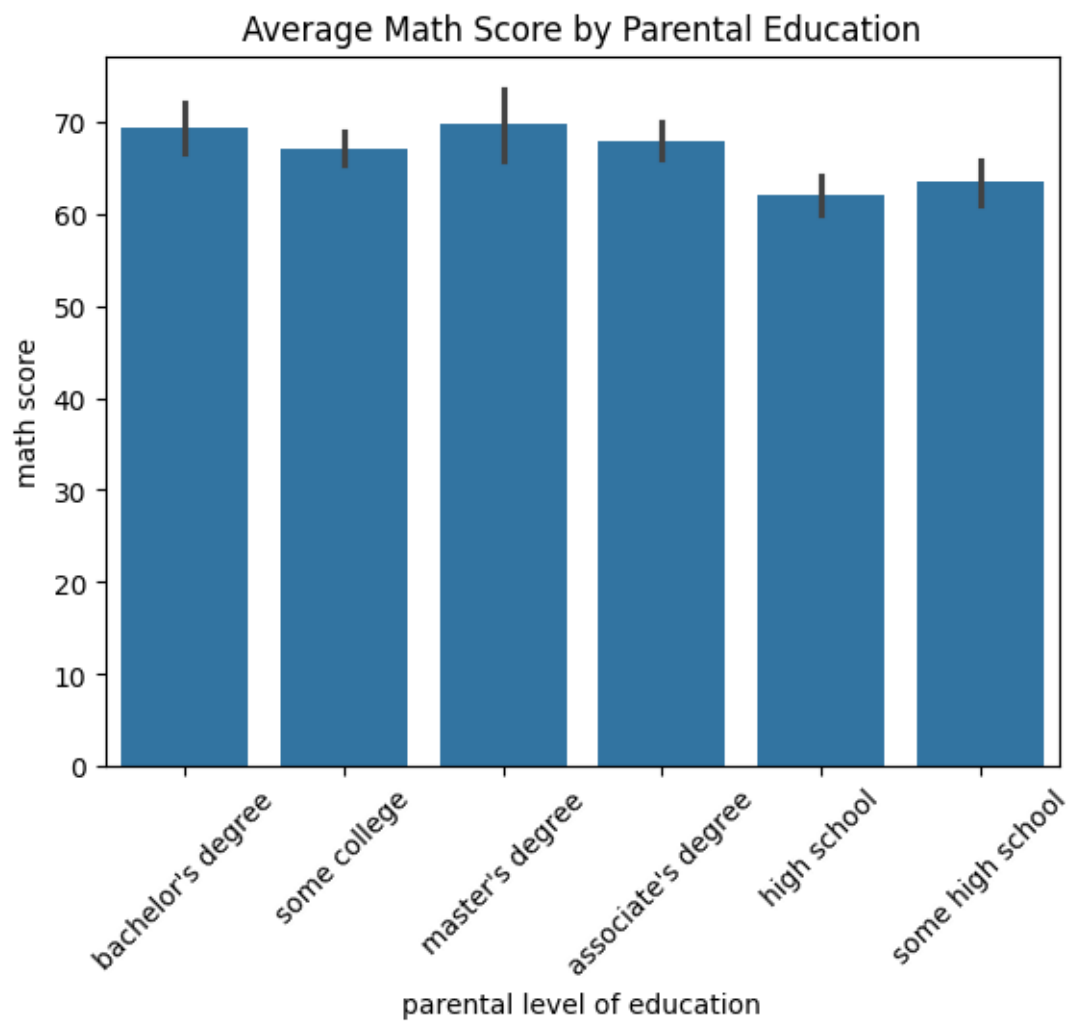
Lunch Type Distribution

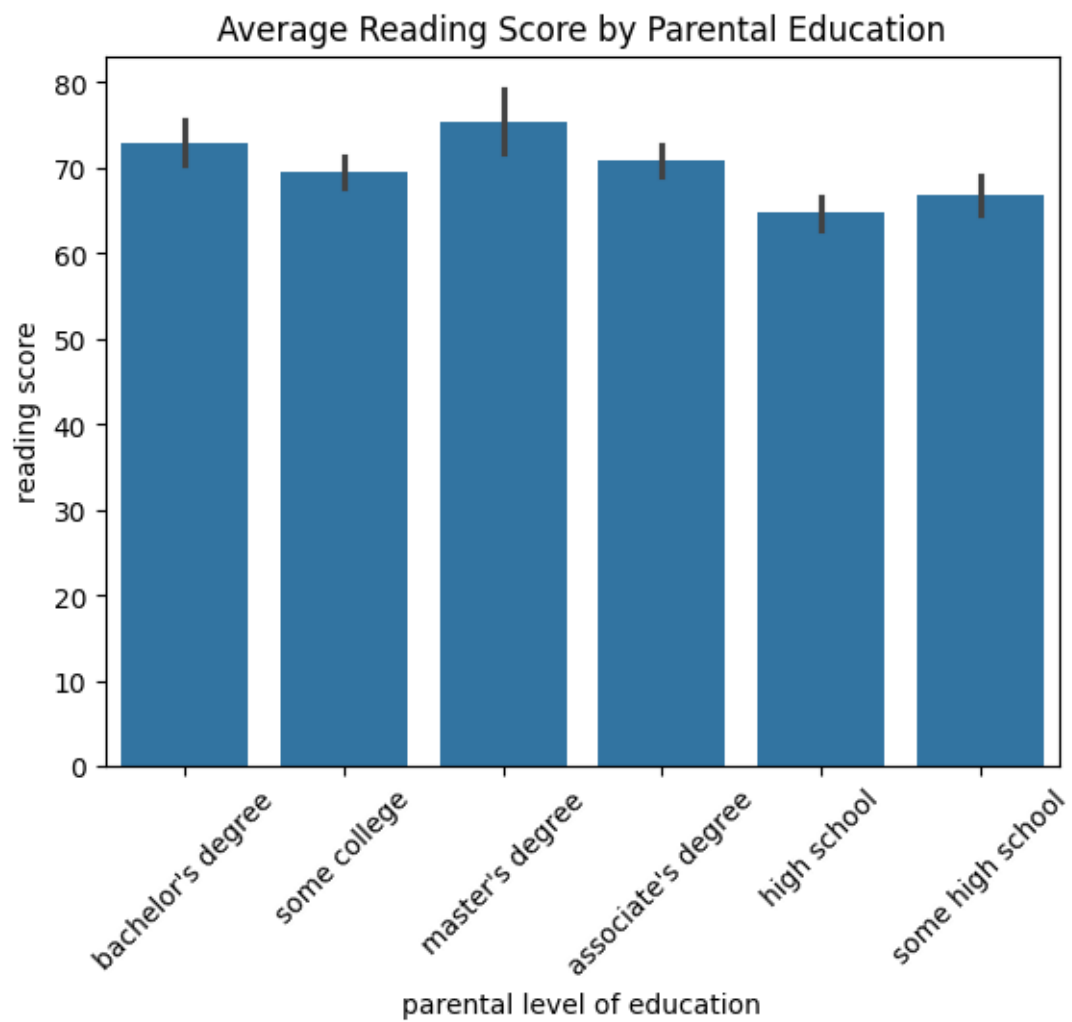


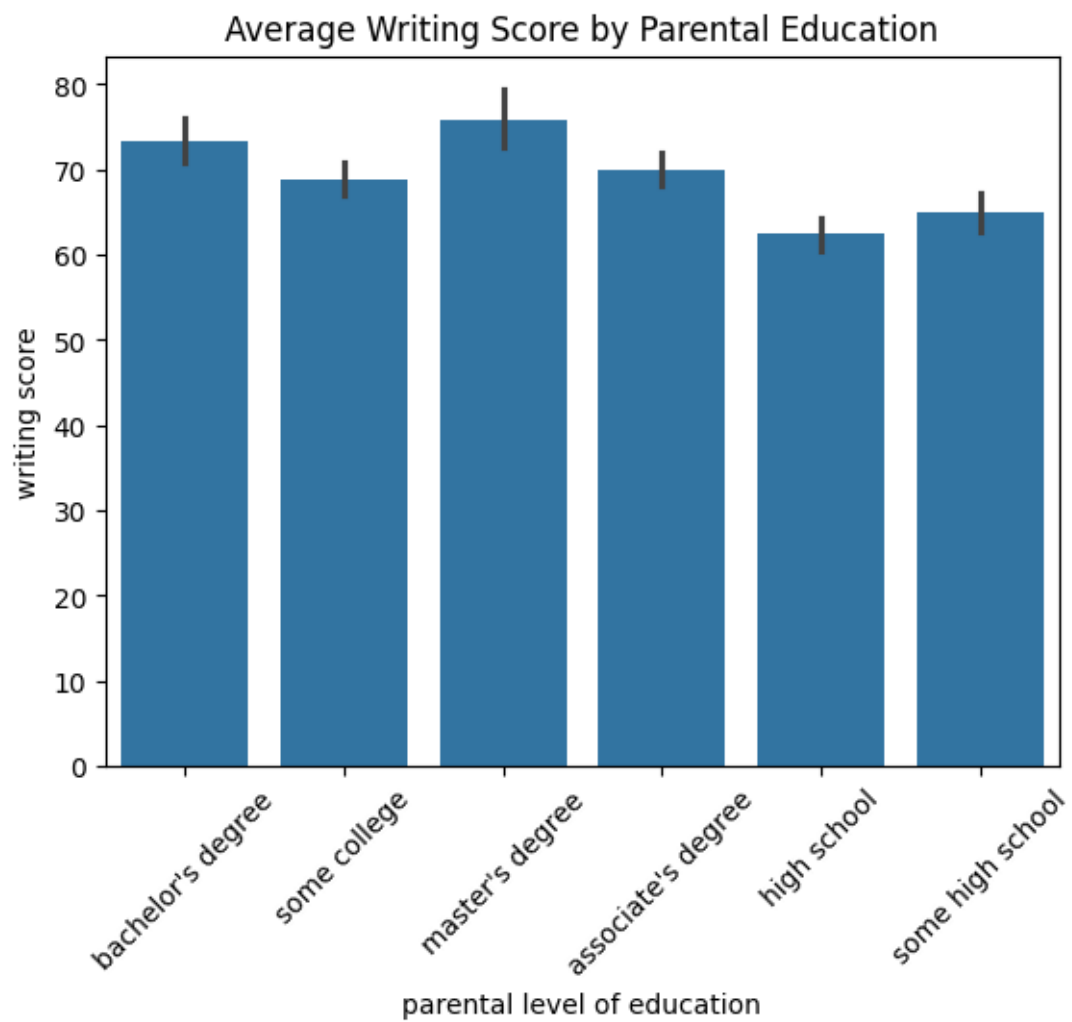
10. Do students who completed the test preparation course score higher on average?



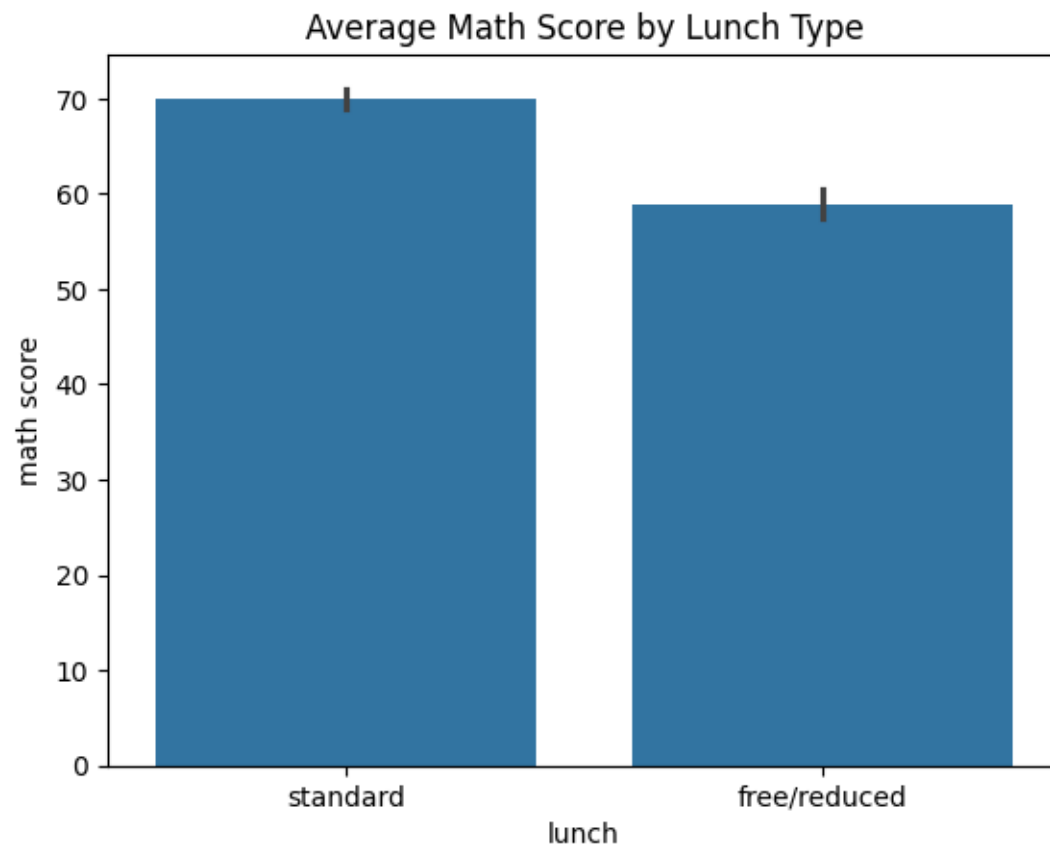
11. How does parental education level affect average math/reading/writing scores?

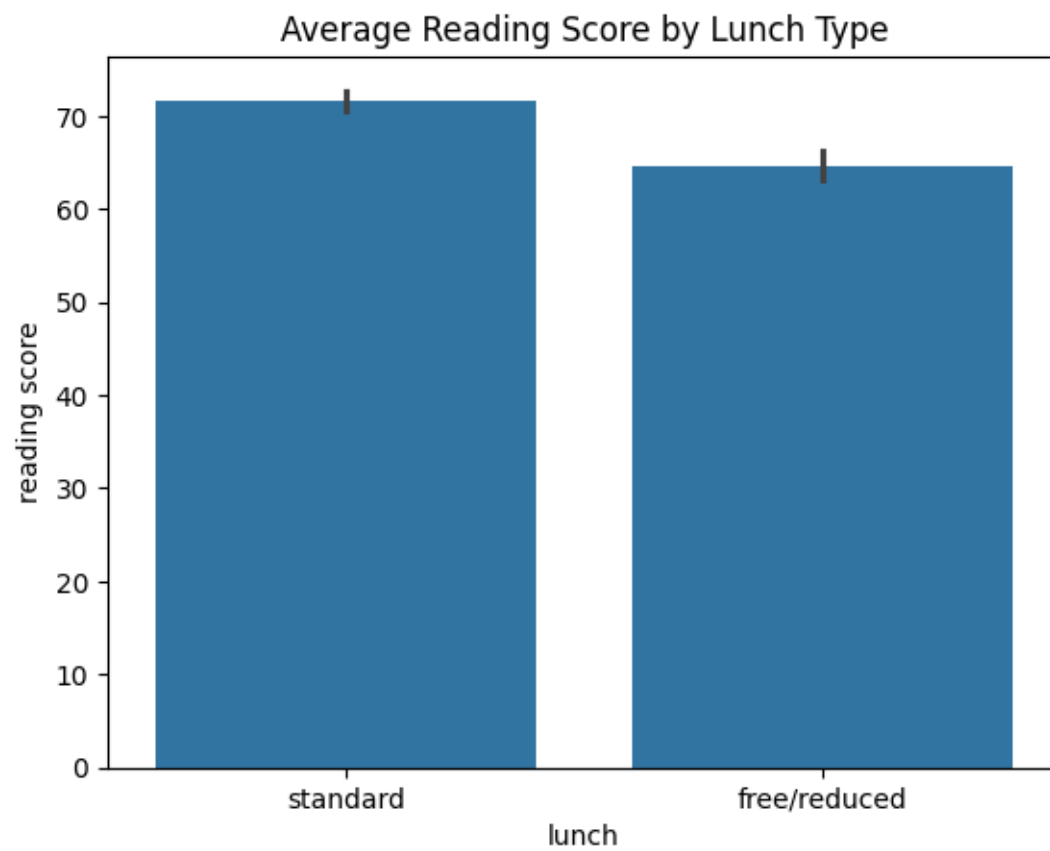


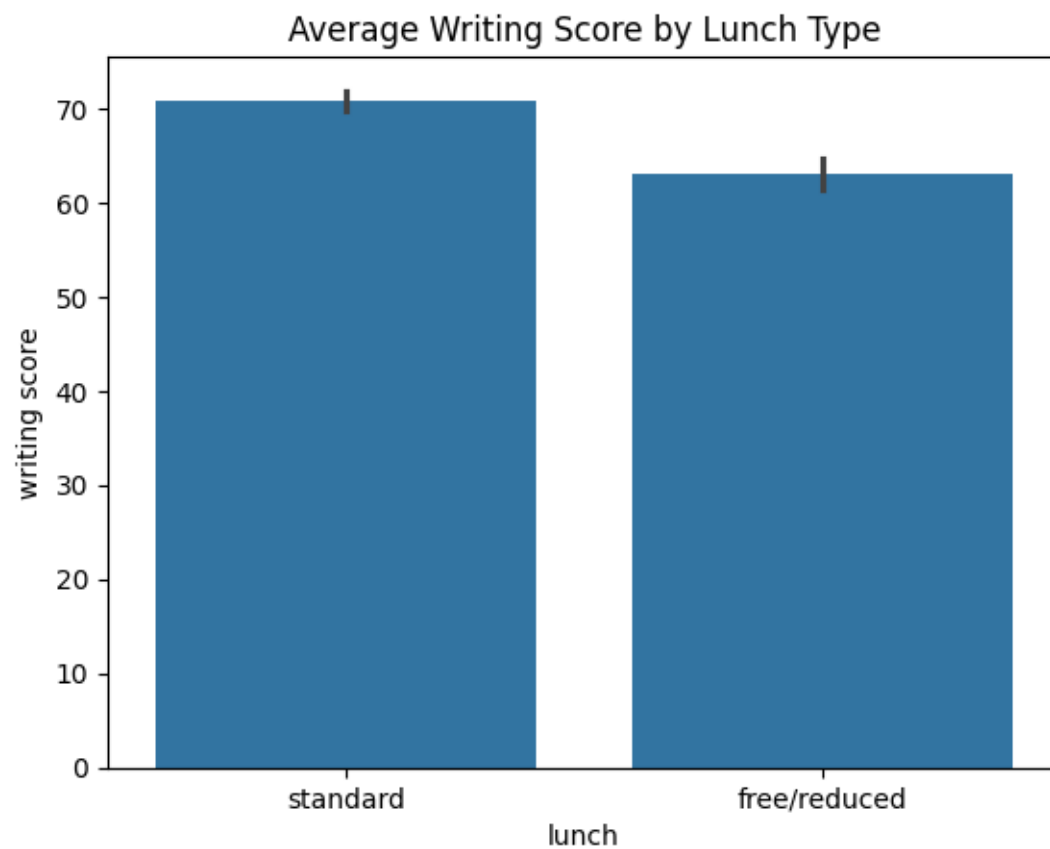




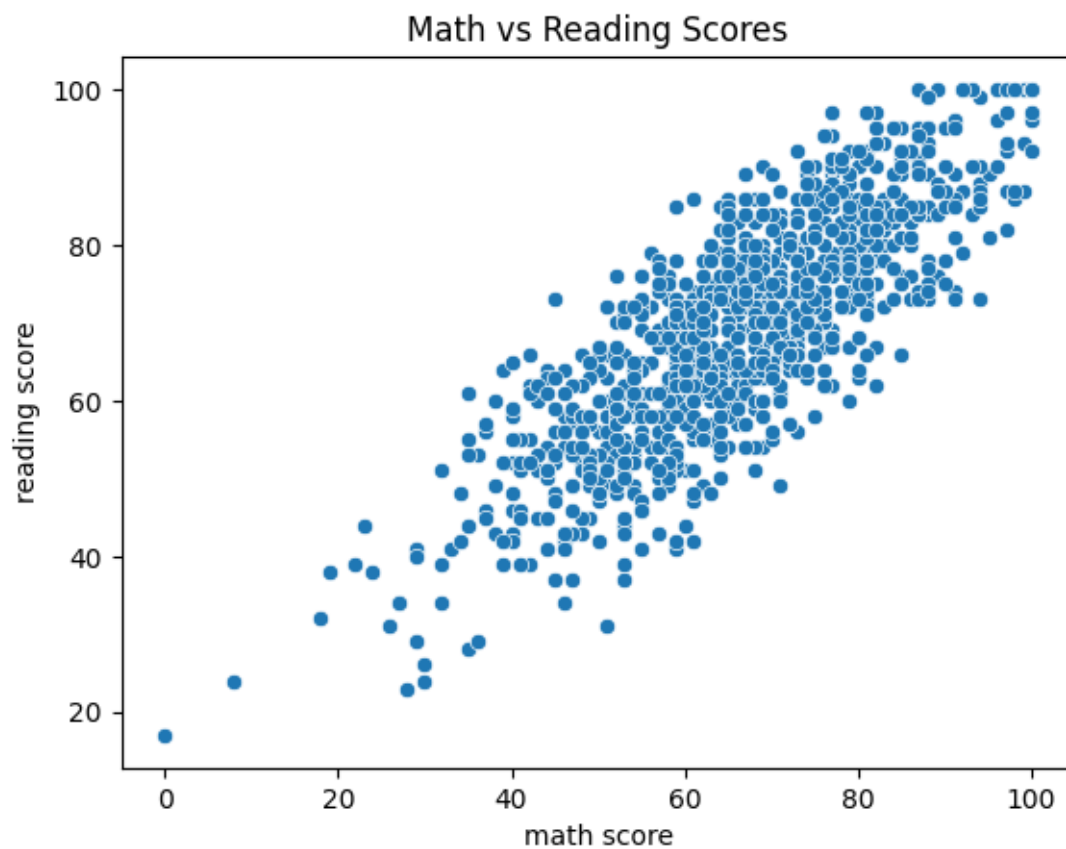
12. How does lunch type affect scores?







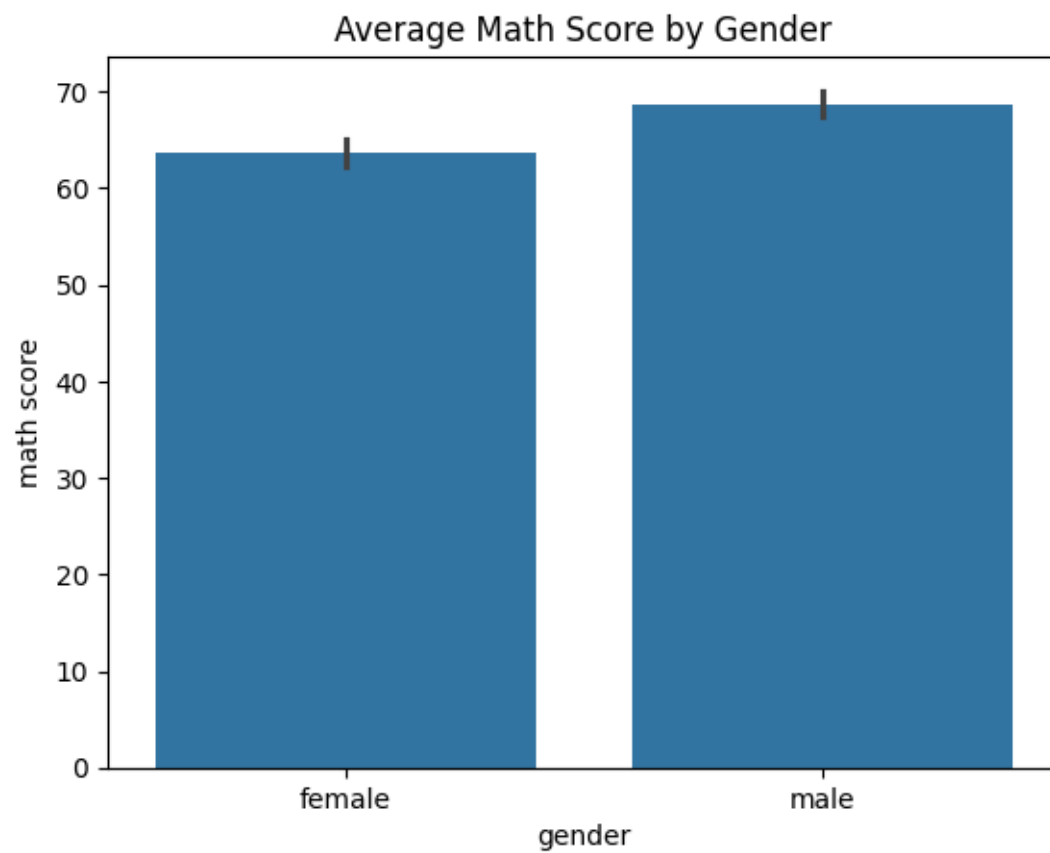
13. What is the relationship between math and reading scores?

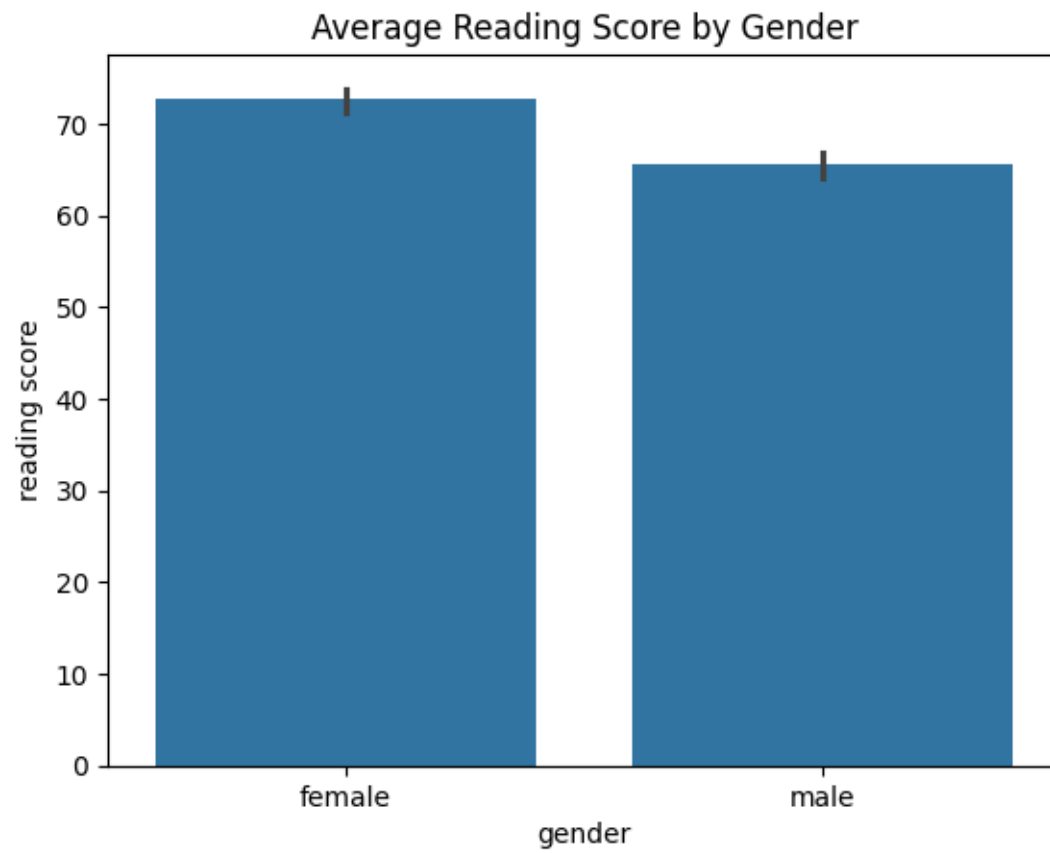


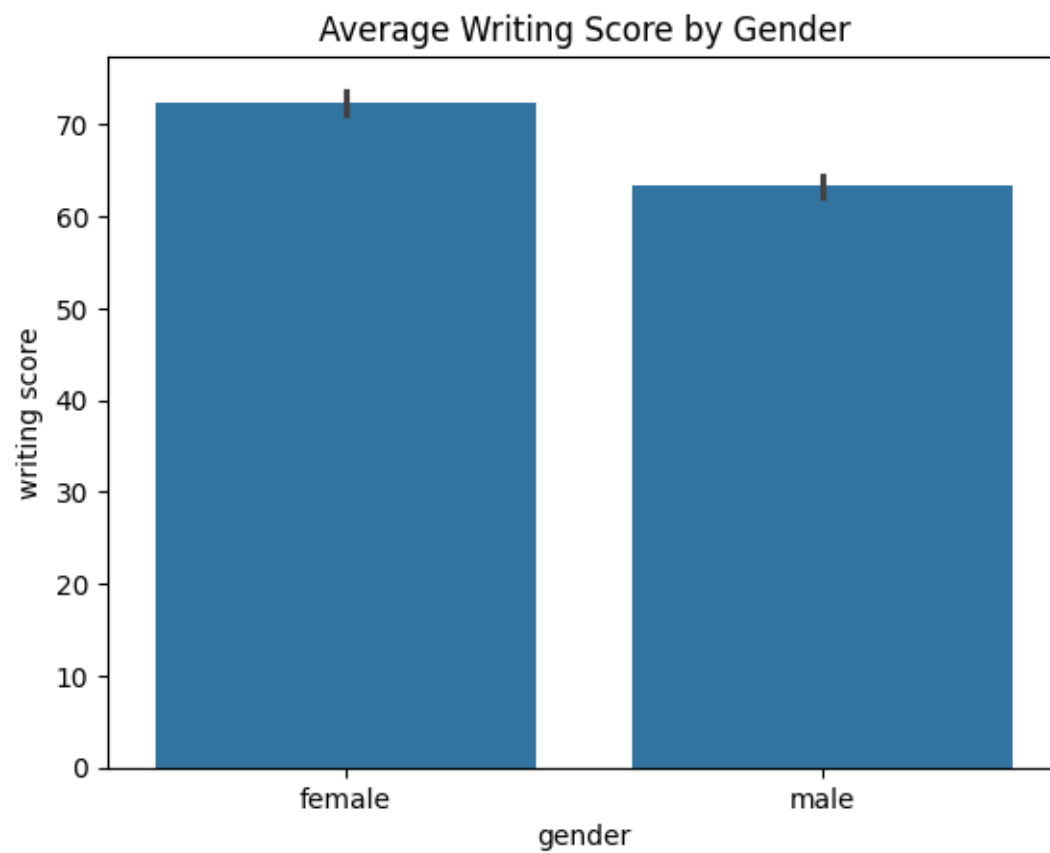
14. What is the relationship between math and reading scores?



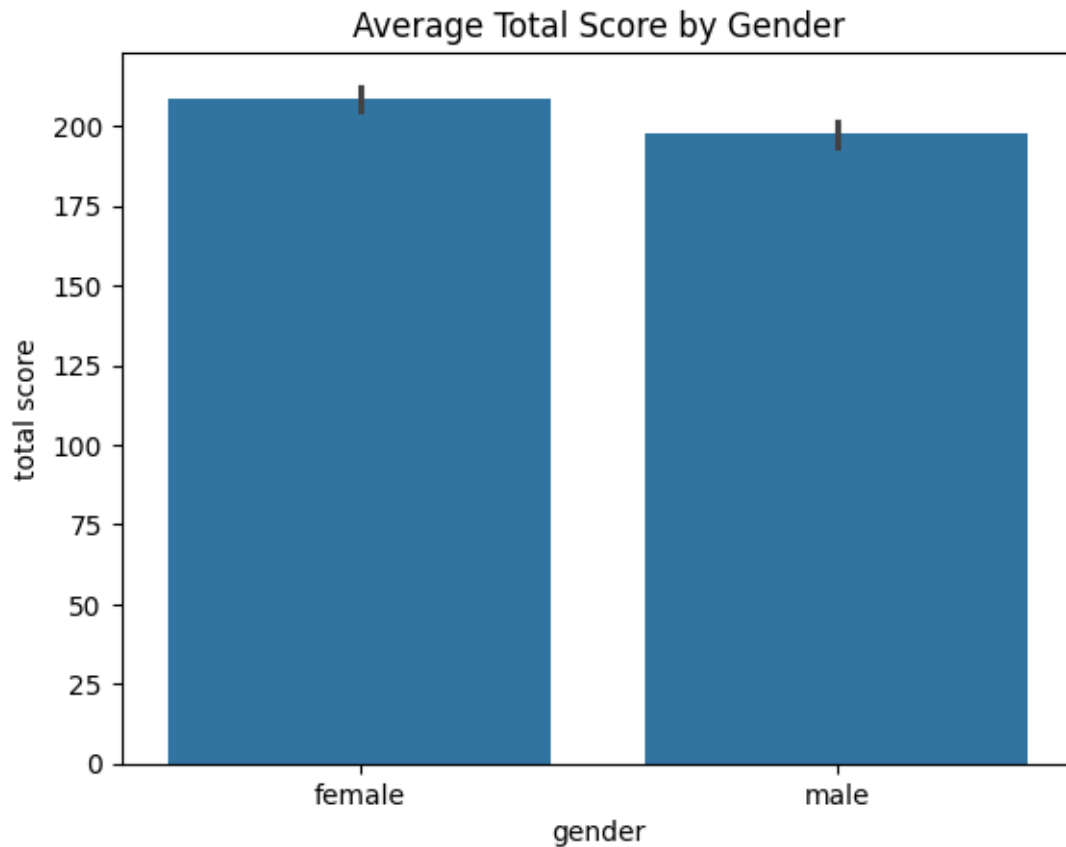
15. Do average scores differ between male and female students?







15. What is the average total score (math + reading + writing) by gender?



VI. SAVE THE CLEANED DATA

```

gender race/ethnicity parental level of education lunch \
0 female group B bachelor's degree standard
1 female group C some college standard
2 female group B master's degree standard
3 male group A associate's degree free/reduced
4 male group C some college standard
.. ...
995 female group E master's degree standard
996 male group C high school free/reduced
997 female group C high school free/reduced
998 female group D some college standard
999 female group D some college free/reduced

```

```

test preparation course math score reading score writing score \
0 none 72 72 74
1 completed 69 90 88
2 none 90 95 93
3 none 47 57 44

```

4	none	76	78	75
..	...	...	...	...
995	completed	88	99	95
996	none	62	55	55
997	completed	59	71	65
998	completed	68	78	77
999	none	77	86	86

	total score
0	218
1	247
2	278
3	148
4	229
..	...
995	282
996	172
997	195
998	223
999	249

[1000 rows x 9 columns]

VII. CONCLUSION

0.1 Insights

1. Female students have slightly higher reading & writing scores.
2. Completing the test preparation course is linked to higher scores.
3. Parental education seems to correlate with student scores.

0.2 Recommendations

- Encourage test prep for all students.
- Provide extra support for students whose parents have lower education levels.